

SOLANA · ZERO-KNOWLEDGE PROOF OF LIFE · SHAMIR + HPKE

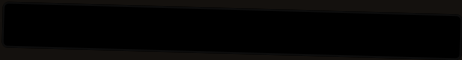
Don't take your keys *to* *the grave.*

SIKRIT is a dead man's switch where proving you're alive doesn't reveal who you are. Your heir inherits your seed phrase only after you fall silent — and only with your guardians' consent.

Muhammad Ghani Nurramdhan · Politeknik Siber dan Sandi Negara
Colosseum Crypto World's Fair 2026

CAPSULE · 9MXQ...T2VA

● alive

Owner 

Heir Fs3k...9hQe

Guardians 2 of 3 must confirm

Last proof heartbeat(R, s) ✓

Signed by owner 



I THE PROBLEM

Self-custody has no next of kin.

2.3–3.7_M BTC

estimated permanently lost — 11-18% of all bitcoin.
Forgotten keys, lost devices, and holders who died
without passing on access.

22.93_M

crypto investors in Indonesia alone (OJK, July 2026).
Most of their families would not know where to begin.

Your family must be able to open your wallet when you're gone —
and must *never* be able to while you're alive.

II THE INSIGHT

Every dead man's switch *broadcasts your pulse.*

A typical on-chain check-in

```
check_in()
  signer: 7xKp... owner wallet ← public
  vault:  PDA("vault", 7xKp...) ← derived from it
  → "7xKp... is alive, active, today"
```

A SIKRIT heartbeat

```
heartbeat(R, s)
  signer:  relayer           ← pays the fee
  capsule: PDA("capsule", P) ←  $P = x \cdot G$ 
  → "whoever knows  $x$  is alive"
```

We reviewed **11 inheritance protocols**. In every one, the check-in points at the owner's wallet — **including the one that already uses a zero-knowledge proof**. A public feed of when a holder is active, and when they stop, is a map for attackers.



III HOW IT WORKS

Seal. Prove. Release.

I

Seal

Your secret is encrypted in your browser under a one-time key. The key is split with Shamir: one share for your heir, one per guardian — each sealed (HPKE, RFC 9180) to a key its holder's wallet signed for.

on-chain: share hashes only

II

Prove you're alive

A Schnorr zero-knowledge proof that you still hold a liveness key that is not your wallet. Verified on-chain, bound to a counter so it works exactly once, sent by any relayer.

≈ 41,000 compute units

III

Release on silence

Miss your interval and anyone may open a claim. A heartbeat cancels it; guardians can veto a false alarm. After the grace period and a guardian quorum, your heir claims — only then do guardians release their shares.

reassembled in the heir's browser



IV PROOF, NOT PROMISES

A proof, not a person.

What the chain sees

heartbeat

tx 4HBw33...jMtYF3

ACCOUNTS (3)

Relayer · fee payer

signer

writable

Es77B...KXHMa

Your capsule (derived from P)

writable

5f7vw...kUvp4

SIKRIT program

FJKqf...Tc45F

INSTRUCTION DATA (72 BYTES)

discriminator

ca683806f0aa3f86

R = k·G

fdf9a9ec24bfee8992e8c454aa222c25ad5e9bffe9953eb94d04c3efebf11b72

s = k + e·x

a53e9fd6b8b23965a893c4b180b33db81df00342612e4b66df2036d8d07abc0e

NOT IN THIS TRANSACTION

Your wallet (Pak Arif)

62DM1G...Ckr1xp

NOT PRESENT

NEVER LEAVES THE DEVICE

liveness secret x

the sealed secret

wallet signature used to derive x

One signature: the relayer that paid the fee. Anyone can read this transaction on any explorer; it carries a zero-knowledge proof, not an identity.

0 of 6

capsule transactions that contain the owner's wallet — re-read from the chain by our end-to-end test after a full inheritance.

41,012 CU

to verify a heartbeat proof, via Solana's curve25519 syscalls.

0.0078 SOL

for 30 years of weekly heartbeats (5,000 lamports each). No token.

V WORKING PRODUCT

One family. Five wallets. *One browser.*

YOUR KEY TO THIS CAPSULE

 3 / 3 shares

Kit verified against the chain: same capsule, same share hashes, holders certified by the heir and guardian wallets registered on-chain.

You hold share #0.

✓ guardian share #1 released to you and authenticated

✓ guardian share #2 released to you and authenticated

Add

RECOVERED SECRET

1 abandon	2 ability	3 able
4 about	5 above	6 absent
7 absorb	8 abstract	9 absurd
10 abuse	11 access	12 accident



Hold to reveal

Clear from screen

Reconstructed in this browser from authenticated shares. Move the funds to a wallet you control.

Sari, the heir · her share + two guardian releases rebuild the seed

• **Alive**

5f7vwT...qkUvp4

YOUR CAPSULE STAYS SEALED FOR

00:59

Proof generated in your browser · relayed by Es77...XHMa



Send ZK heartbeat

Pak Arif, the owner · one click: proof generated in the browser, sent by a relayer

• **Claim pending**

capsule 5f7vwT...qkUvp4

heir: Sari · you are guardian #2

CLAIM OPEN — GRACE PERIOD ENDS IN

00:56

interval 1m · grace 1m · guardians 2/2 confirmed · 1 heartbeat

Veto — the owner is alive

Confirmed

tx 2vu378...V7orh5 ↗

Budi & Dewi, guardians · confirm the claim, or veto a false alarm

VI UNDER THE HOOD

Built like it will be attacked.

```
// Fiat-Shamir, mirrored byte-for-byte TS ↔ Rust
e = SHA-512("SIKRIT:liveness:v1" || program
           || capsule || P || R || nonce) mod ℓ

// on-chain, sol_curve25519 syscalls
accept ↔ s·G - e·P = R
```

The counter makes every proof single-use; domain, program and capsule bind it to one purpose. A pure-Rust verifier blew past 1.4M CU — the syscall path costs 41k.

- ✓ HPKE RFC 9180 · X25519 · HKDF-SHA256 · ChaCha20-Poly1305 · official test vectors
- ✓ Shamir over $GF(2^8)$ · audited library (Cure53, Zellic) · only a random key is split
- ✓ Inbox certificates · a swapped encryption key is rejected, at sealing and at release
- ✓ Self-audit · 17 findings (16 fixed, 1 mitigated) · residual risks R1-R16 published
- ✓ 62 tests + 10 Rust unit tests · plus a browser end-to-end run of the whole story

HONEST LIMITS Heartbeat times and heir/guardian addresses are public. Enough colluding guardians could open a kit early. Not post-quantum yet. All on the roadmap.

VII LANDSCAPE

Inheritance exists. *Private liveness doesn't.*

PROJECT	CHAIN	IS THE CHECK-IN LINKED TO THE OWNER?
DeathClock	Solana	Yes — its Groth16 heartbeat proof commits to the owner's key (~183k CU)
Dead Man's Vault	Solana	Yes — vault derived from the owner's wallet
Eternal	EVM	Yes — every owner action is the proof of life (heirs are hidden, the owner is not)
Sarcophagus	Base + Arweave	Yes — owner re-wraps from their wallet; relies on the SARCO token
LastSats · Heirloom	Stacks	Yes — owner check-in transactions
Kaspa Safe	Kaspa	Yes — periodic on-chain check-in
Serenity Shield · Deadhand	Secret · server	Trusted party — a TEE network or a server sees every check-in
SIKRIT	Solana	No — a zero-knowledge proof from a non-wallet key, sent by a relayer

VIII BUSINESS

Open core. *No token.*

B2B2C

Wallet integrations

An "Inheritance" tab inside Solana wallets, built on our dependency-light SDK. Retention for them, distribution for us.

SUBSCRIPTION

Watcher & relay

\$2-5 / month: heartbeat reminders, an alarm the moment a claim opens, relaying with an SLA. No lock-in — anyone can relay.

SERVICES

Professional guardians

Notaries and estate planners as guardians — a natural fit for Indonesia's *akta wasiat*. They confirm; they can never open alone.

PROTOCOL

Release fee

Optional, paid once — only when an inheritance actually happens. Heartbeats stay near-free forever.

Indonesia first → crypto communities, campuses, notaries → self-custody users worldwide through their wallets.

Why no token: a dead man's switch must outlive market cycles. SIKRIT's program and SDK stay open source; revenue comes from services around them, not from speculation.



IX TEAM & NEXT

Built by a cryptographer-in-training.

Muhammad Ghani Nurramdhan



Founder · 4th-year student, Cryptographic Software Engineering at Politeknik Siber dan Sandi Negara — Indonesia's state polytechnic for cyber security and cryptography.

NEXT · V1.1

- Watcher alerts when a claim opens
- Origin-bound key derivation
- Ledger support
- External audit → mainnet

THEN · V2

- Heartbeats inside an anonymity set — hide *which* capsule
- Heirs & guardians as hash commitments
- Post-quantum hybrid KEM (X-Wing)

Prove you're alive.
Reveal nothing else.

code github.com/Darkside0908/SIKRIT
program FJKqf...Tc45F
license MIT · open source

